

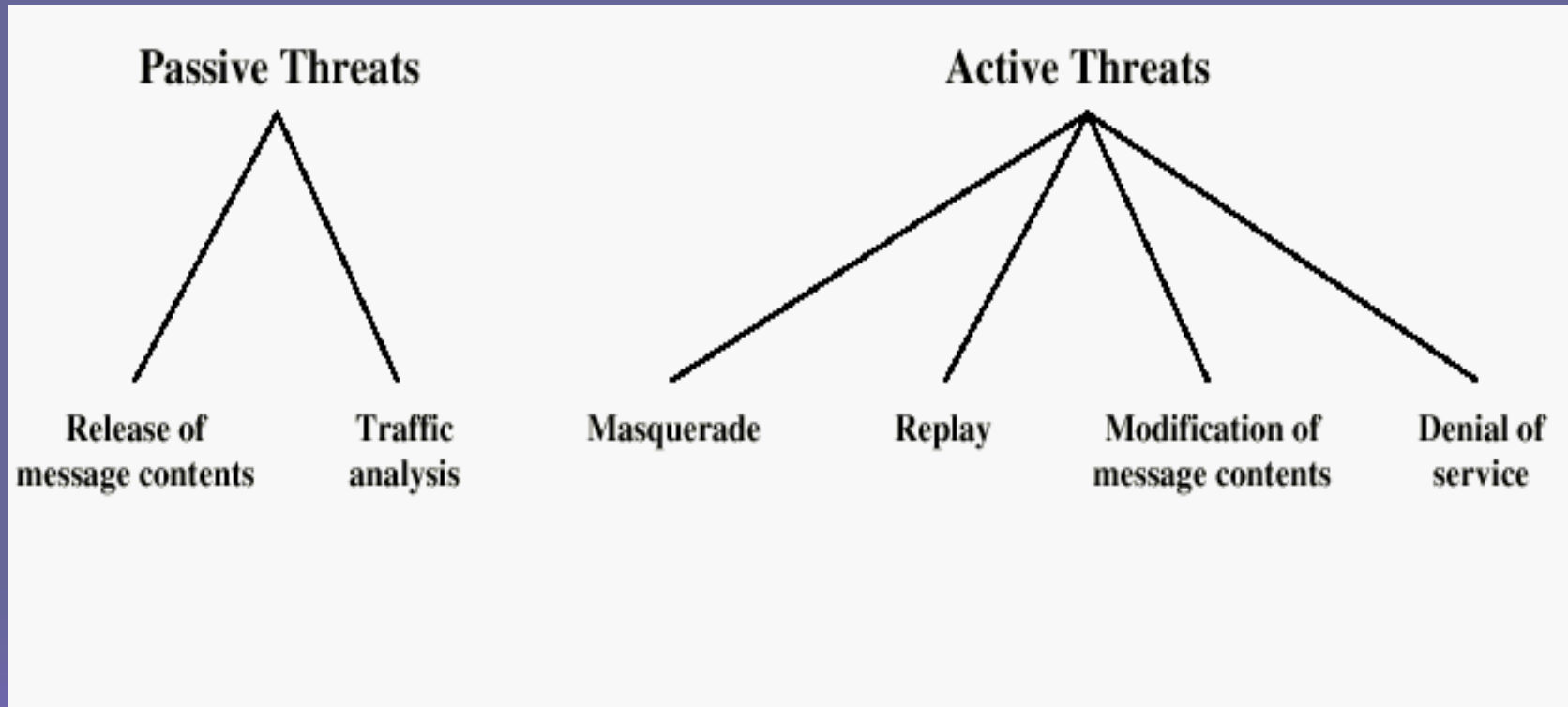


Sir Syed University of Engineering & Technology, Karachi

Cryptography and Network Security CS-317

Department of Computer Science

Security Attacks

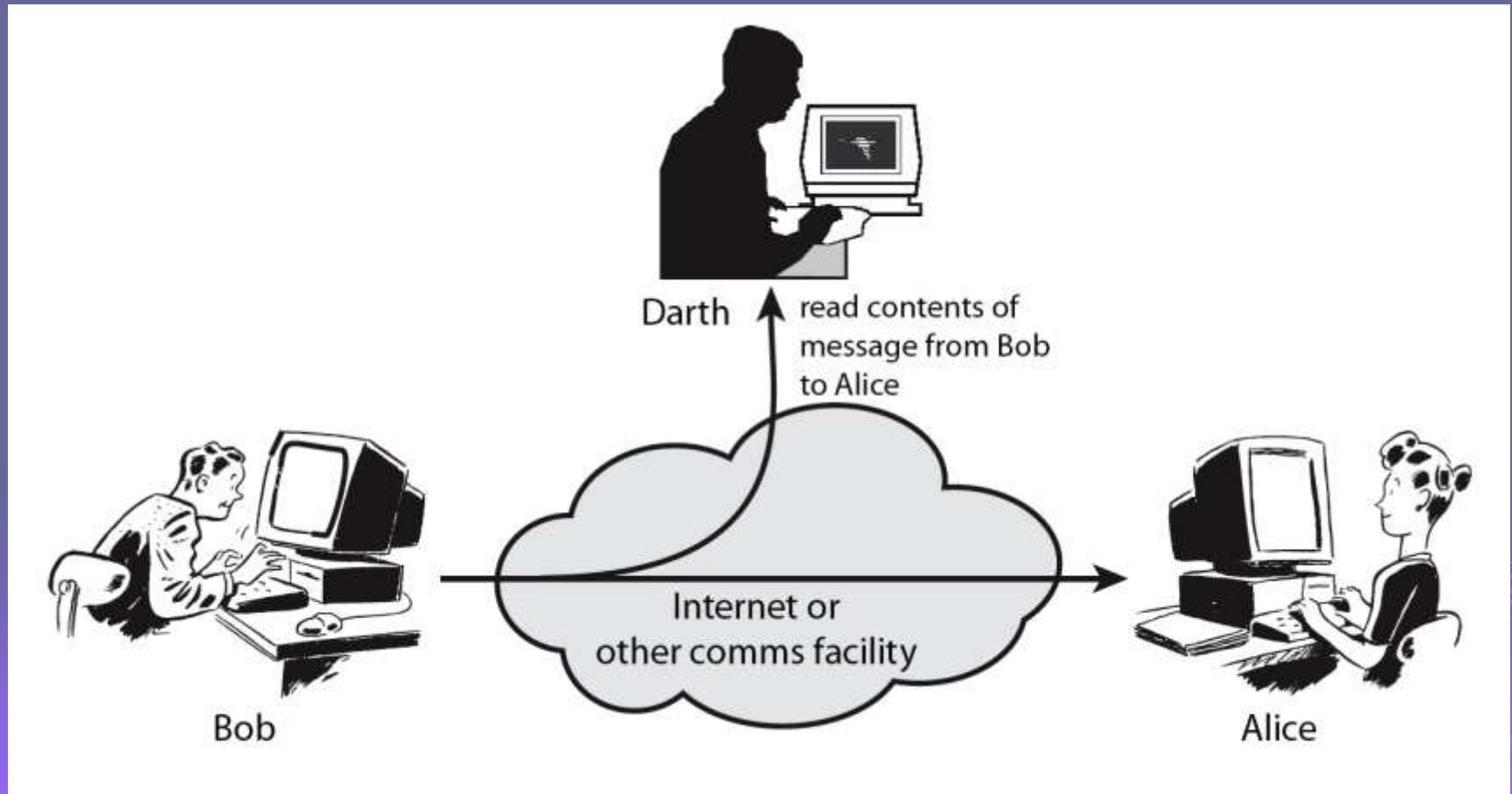


Learning Assignment: In the notes (below)

Security Attacks

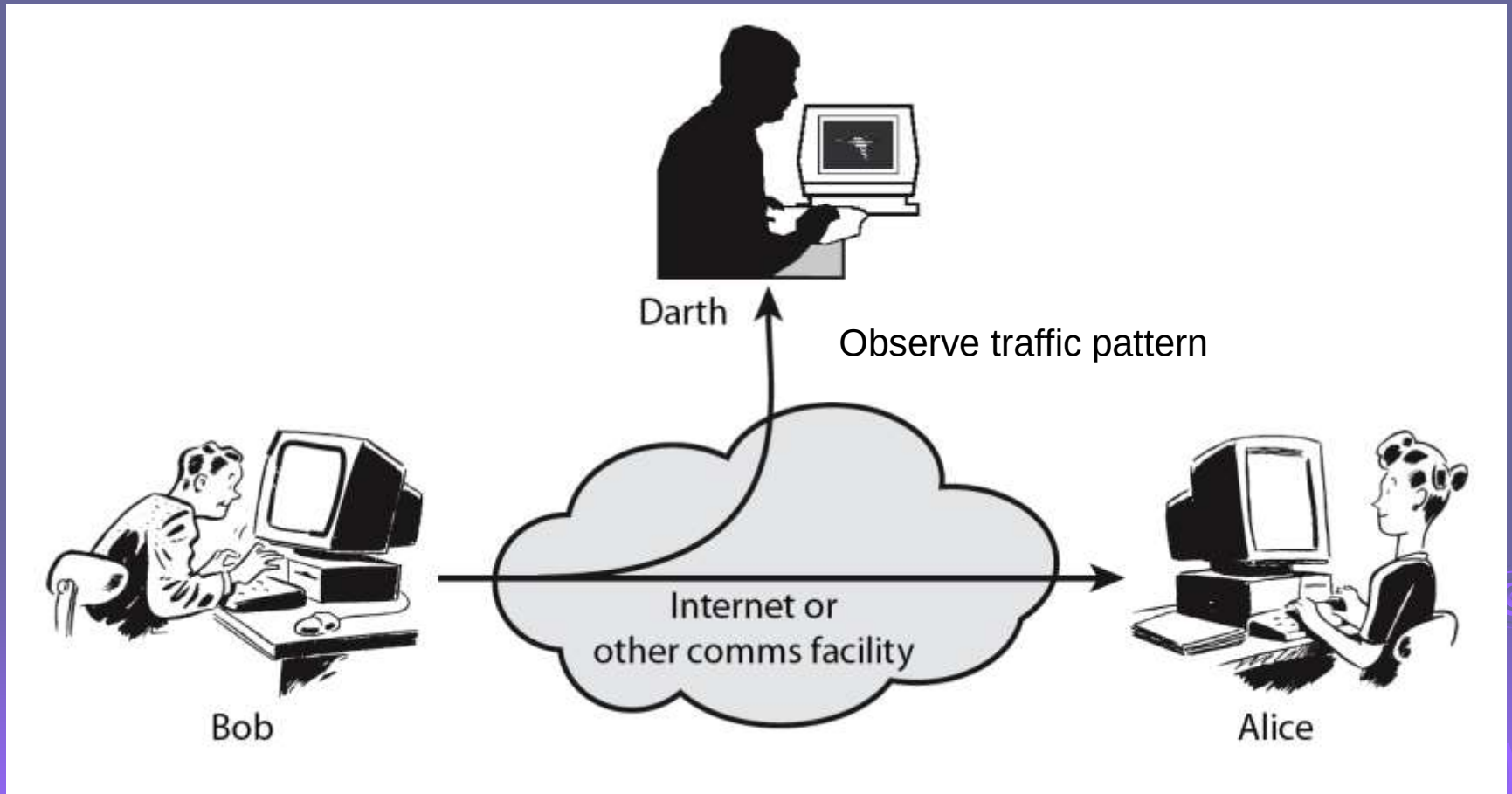
- **Passive attacks:** are in the nature of eavesdropping on, or monitoring of, transmissions
- Goal of the opponent is to obtain information that is being transmitted
- **Active attacks:** involve some modification of the data stream or the creation of a false stream

Passive Attack - Interception

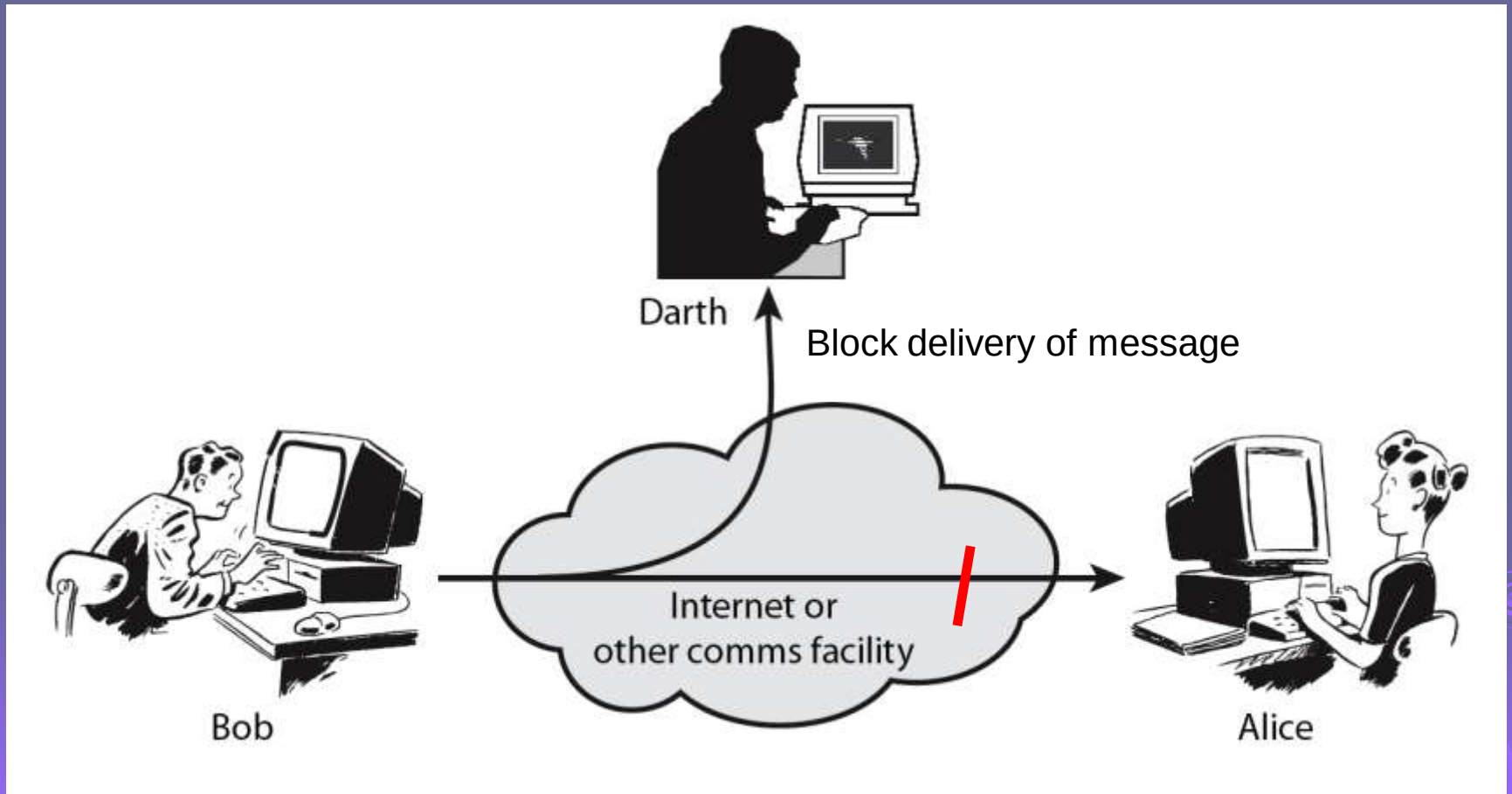


- This is an attack on confidentiality

Passive Attack: Traffic Analysis

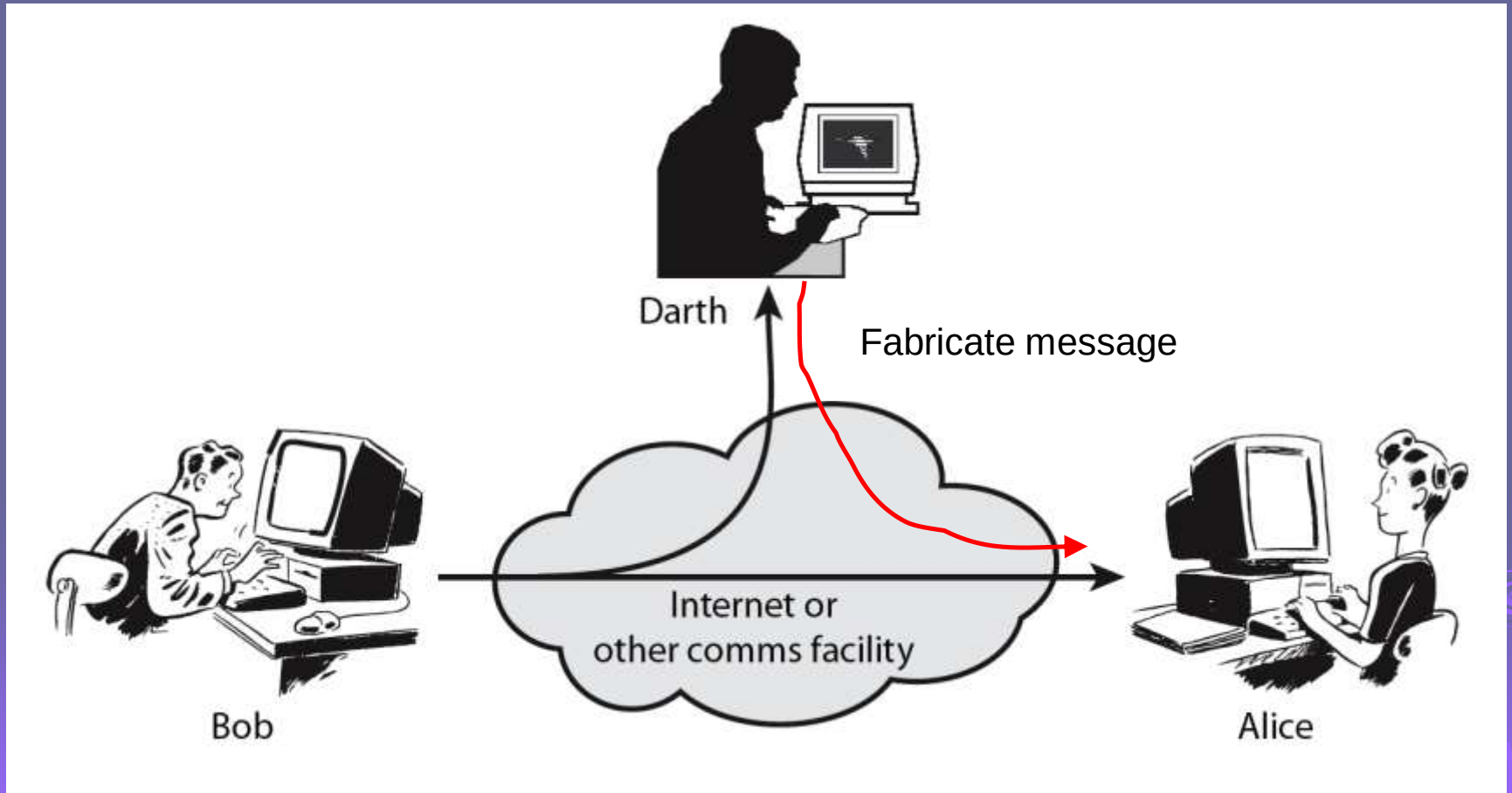


Active Attack: Interruption



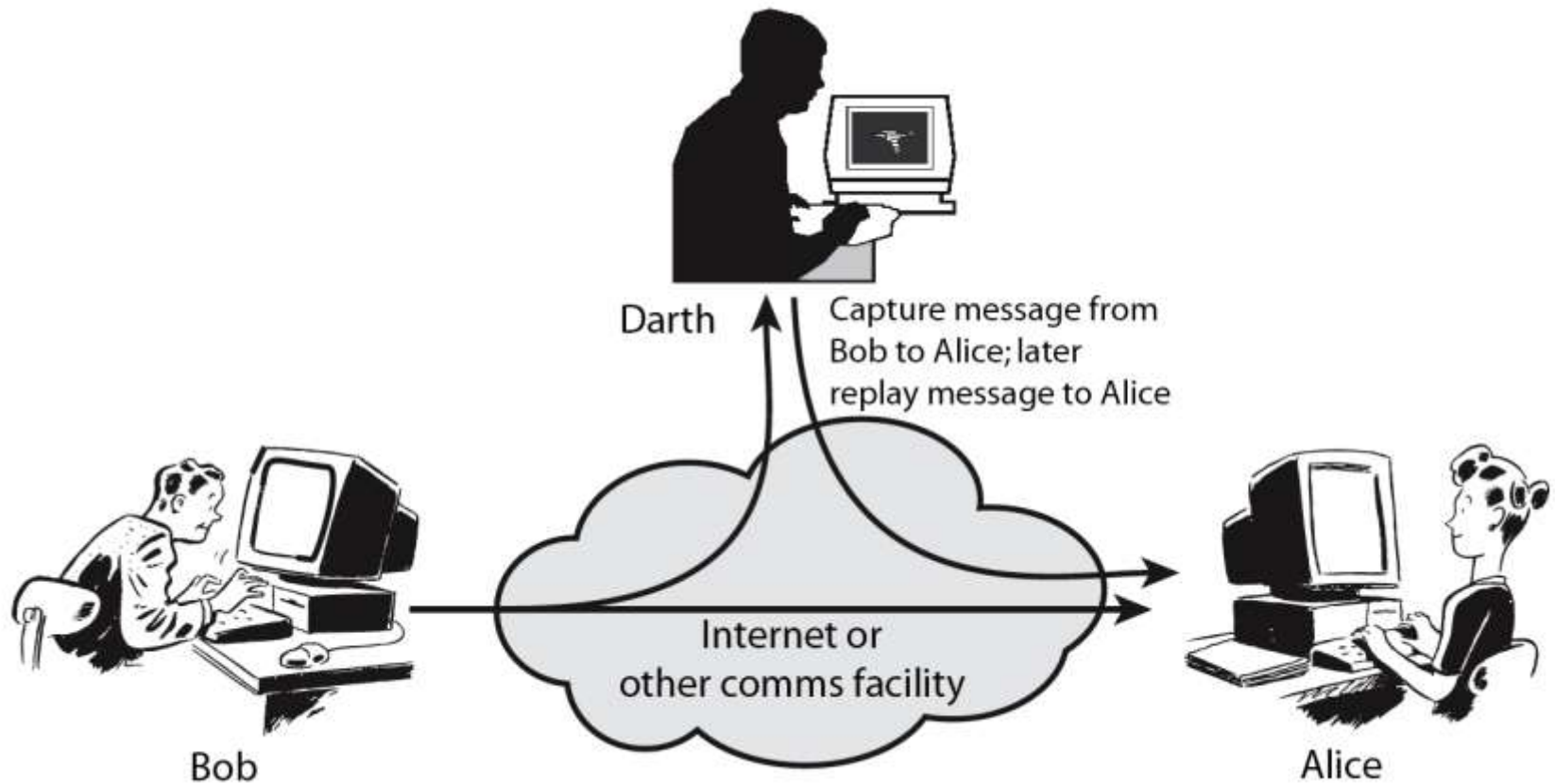
- This is an attack on availability

Active Attack: Fabrication

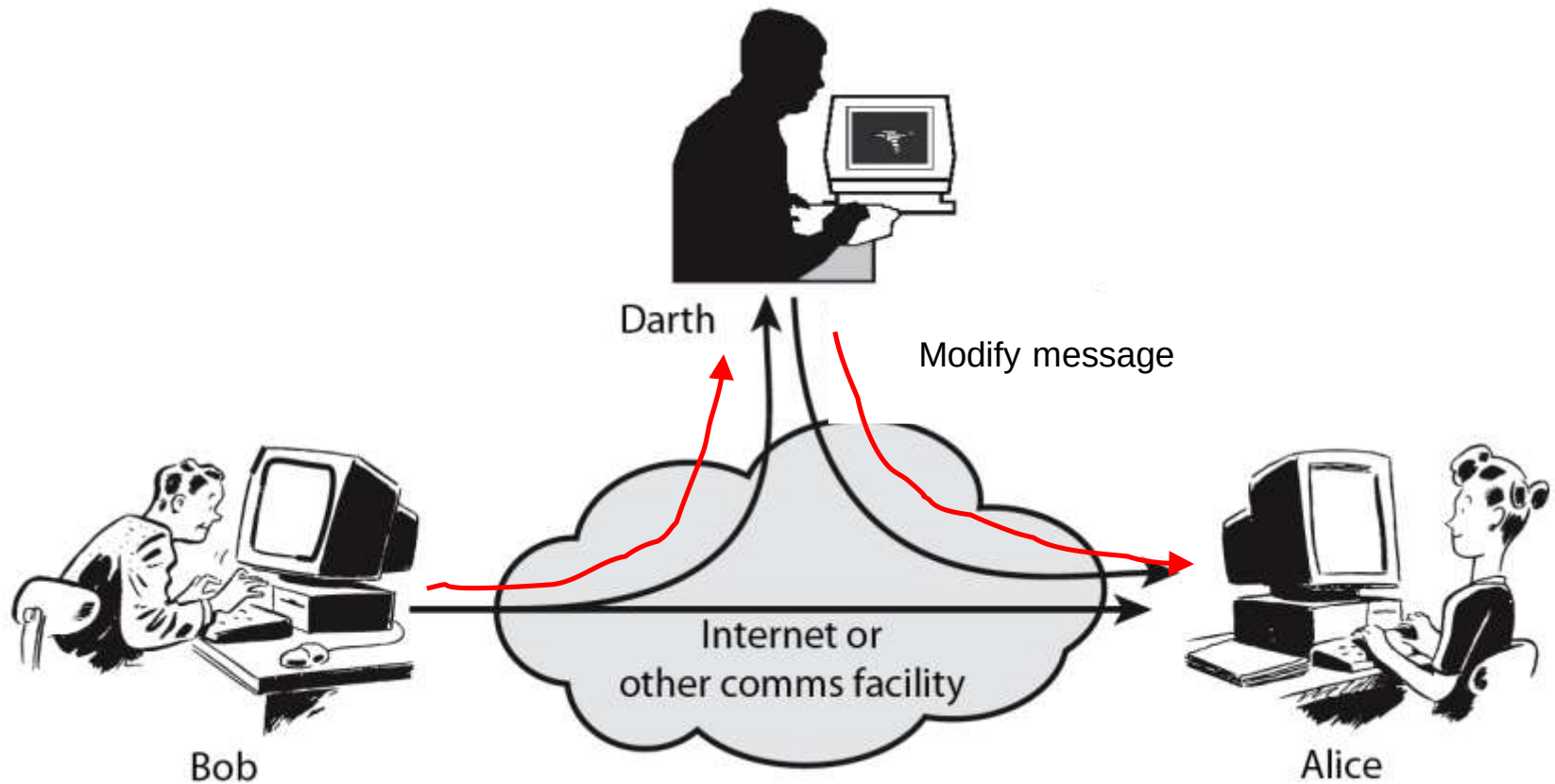


- This is an attack on authenticity

Active Attack: Replay



Active Attack: Modification



This is an attack on integrity

Handling Attacks

- Passive attacks – focus on Prevention
 - Easy to stop
 - Hard to detect
- Active attacks – focus on Detection and Recovery
 - Hard to stop
 - Easy to detect

How to do an Attack

- ◆ The following describes some of the ways through which any information systems can be attacked.
 - a. Social Engineering
 - b. Viruses / Trojan Horses / Worms
 - c. Denial of Service (DoS)
 - d. IP Spoofing
 - e. Replay Attack

Social Engineering

- ◆ Social engineering is a technique used by attackers to gain system access or information by exploiting the basic human instinct to be helpful.
- ◆ Social engineering exploits are successful because the targeted enterprise lacks an awareness program to educate employees of their security-related duties and responsibilities.
- ◆ A simple phone call by an intruder to a target posing as a network support guy and asking her username and password to rectify a non existing problem.

Viruses / Trojan Horses / Worms

- ◆ Virus is malicious code that can plant itself into operating systems and programs and modify them.
- ◆ Trojan-horse is a virus that is hidden inside a legitimate software. Once the software is installed or downloaded, the malicious code does its thing.
- ◆ Worm is industry nomenclature for a self-contained program that will replicate itself across a network, infecting each server and workstation it can access.

Denial of Service (DoS)

- ◆ An attack that targets resources within the network with the intention of reserving resource and keeping legitimate users from gaining access.
- ◆ During a *SYN* attack, an enemy workstation will generate a flood of session requests using bogus IP addresses.
- ◆ The target server begins reserving resources for each request while waiting for the completion of the TCP/IP handshake process.
- ◆ The expected reply from the enemy workstation never comes but the reserved resources results in the denial of service for the legitimate user.

IP Spoofing

- ◆ IP spoofing is accomplished when an outside hacker uses a discovered IP address to gain access of a trusted environment from outside the network.
- ◆ A hacker can obtain a valid IP address in a variety of ways such as social engineering.



Replay Attack

- ◆ A replay attack occurs when a hacker intercepts a communication between two parties and replays the message.
- ◆ For instance, a hacker might intercept a credit card transaction between a consumer and a Web site. The hacker then replays the transaction multiple times resulting in multiple debits to the consumers credit account.